

Unit 1

1. Choose the correct option in terms of Issues related to professional responsibility

- a) Confidentiality
- b) Intellectual property rights
- c) Both Confidentiality & Intellectual property rights
- d) Managing Client Relationships

Answer: c

2. “Software engineers should not use their technical skills to *misuse* other people’s computers.” Here the term *misuse* refers to:

- a) Unauthorized access to computer material
- b) Unauthorized modification of computer material
- c) Dissemination of viruses or other malware
- d) All of the mentioned

Answer: d

3. Explain what is meant by *PRODUCT* with reference to one of the eight principles as per the ACM/IEEE Code of Ethics ?

- a) The product should be easy to use
- b) Software engineers shall ensure that their products and related modifications meet the highest professional standards possible
- c) Software engineers shall ensure that their products and related modifications satisfy the client
- d) It means that the product designed /created should be easily available

Answer: b

4. Identify an ethical dilemma from the situations mentioned below:

- a) Your employer releases a safety-critical system without finishing the testing of the system
- b) Refusing to undertake a project
- c) Agreement in principle with the policies of senior management
- d) All of the mentioned

Answer: a

5. Identify the correct statement: “Software engineers shall

- a) act in a manner that is in the best interests of his expertise and favour.”
- b) act consistently with the public interest.”
- c) ensure that their products only meet the SRS.”
- d) all of the mentioned

Answer: b

6. Select the incorrect statement: “Software engineers should
- a) not knowingly accept work that is outside your competence.”
 - b) not use your technical skills to misuse other people’s computers.”
 - c) be dependent on their colleagues.”
 - d) maintain integrity and independence in their professional judgment.”

Answer: c

7. Efficiency in a software product does not include _____
- a) responsiveness
 - b) licensing
 - c) memory utilization
 - d) processing time

Answer: b

8. As per an IBM report, “31% of the project get cancelled before they are completed, 53% overrun their cost estimates by an average of 189% and for every 100 projects, there are 94 restarts”.What is the reason for these statistics ?
- a) Lack of adequate training in software engineering
 - b) Lack of software ethics and understanding
 - c) Management issues in the company
 - d) All of the mentioned

Answer: a

9. Which of the following property does not correspond to a good Software Requirements Specification (SRS) ?
- a) Verifiable
 - b) Ambiguous
 - c) Complete
 - d) Traceable

Answer: b

10. Which one of the following is a functional requirement ?

- a) Maintainability
- b) Portability
- c) Robustness
- d) Operations of the system

Answer: d

11. _____ requirements capture the intended behavior of the system.

- a) Functional
- b) Non-functional
- c) Emergent
- d) Quality

Answer: a

12. The SRS document is also known as _____ specification.

- a) black-box
- b) white-box
- c) grey-box
- d) green-box

Answer: a

13. Requirements should specify _____ but not 'how'.

- a) 'what'
- b) 'how'
- c) 'why'
- d) 'which'

Answer: a

14. Requirements Analysis is an _____ Process.

- a) Iterative
- b) Non-Iterative
- c) Sequential
- d) Unidirectional

Answer: a

15. SDLC stands for

- a) Software Development Life Cycle
- b) System Development Life cycle
- c) Software Design Life Cycle
- d) System Design Life Cycle

Answer: a

16. The spiral model was originally proposed by

- a) IBM
- b) Barry Boehm
- c) Pressman

d) Royce
Answer: b

17. Compilers, Editors software come under which type of software?

- a. System software
- b. Application software
- c. Scientific software
- d. Business Software

Answer: a

18. Which of the following is not defined in a good Software Requirement Specification (SRS) document?

- a. Functional Requirement.
- b. Nonfunctional Requirement.
- c. Goals of implementation.
- d. Algorithm for software implementation.

Answer: d

19. Requirement engineering process includes which of these steps?

- a. Feasibility study
- b. Requirement Gathering
- c. Software Requirement specification & Validation
- d. All choices are correct

Answer: d

20. If requirements are easily understandable and defined then which model is best suited?

- a. Spiral model
- b. Waterfall model
- c. Prototyping model
- d. Incremental

Answer: b

21. Software is defined as ____.

- a. Instructions
- b. Data Structures
- c. Documents
- d. All choices are correct

Answer: d

22. What is the meaning of requirement elicitation in software engineering?

- a. Gathering of requirement.
- b. Understanding of requirement.
- c. Getting the requirements from client.
- d. All choices are

correct

Answer: d

23. Which is not a step of Requirement Engineering?

- a. Requirements elicitation
- b. Requirements analysis
- c. Requirements design
- d. Requirements documentation

Answer: c

24. Find out which phase is not available in SDLC?

- a. Coding
- b. Testing
- c. Maintenance
- d. Abstraction

Answer: d

25. Which of the following is/are considered stakeholder in the software process?

- a. Customers
- b. End-users
- c. Project managers
- d. All choices are correct

Answer: d

26. The process together the software requirements from Client, Analyze and Document is known as _ .

- a. Requirement engineering process
- b. Requirement elicitation process
- c. User interface requirements
- d. Software system analyst

Answer: a

27. For the best Software model suitable for the project, in which of the phase the developers decide a roadmap for project plan?

- a. Software Design
- b. System Analysis
- c. Coding
- d. Testing

Answer: b

28. Which of these primary objectives have to be achieved for the requirement model?

- a. To describe what the customer requires
- b. To establish a basis for the creation of a software design
- c. To define a set of requirements that can be validated
- d. All choices are correct

Answer: d

29. Design phase is followed by _____.

- a. Coding
- b. Testing
- c. Maintenance
- d. Analysis

Answer: a

30. Project risk factor is considered in which model?

- a. Spiral model.
- b. Waterfall model.
- c. Prototyping model
- d. Incremental

Answer: a

31. Abbreviate the term SRS.

- a. Software Requirement Specification
- b. Software Refining Solution
- c. Software Resource Source
- d. Software Service

Answer: a

32. What are the major activities of the spiral model of software engineering?

- (a) Planning, Risk Analysis, Engineering, Customer Evaluation
- (b) Defining, Prototyping, Testing, Delivery
- (c) Requirements
- (d) Quick Design, Build Prototype, Evaluate Prototype, Refine Prototype

Answer: a

33. Symptoms of the software crisis would include

- (a) Software delivered behind schedule
- (b) Software exceeding cost estimate
- (c) Unreliable
- (d) All choices are correct

Answer: d

34. An SRS

- a. establishes the basis for agreement between client and the supplier.
- b. provides a reference for validation of the final product.
- c. is a prerequisite to high quality software.
- d. All choices are correct

Answer: d

35. Main aim of software engineering is to produce

- a. program
- b. software
- c. within budget
- d. software within budget in the given schedule

Answer: d

36. Reliability of software is directly dependent on

- a. quality of the design
- b. number of errors present
- c. software engineers experience
- d. user requirement

Answer: b

37. Working software is not available until late in the process in

- a. Waterfall model
- b. Prototyping model
- c. Incremental model
- d. Evolutionary Development model

Answer: a

38. The major shortcoming of waterfall model is

- a. the difficulty in accommodating changes after requirement analysis.
- b. the difficult in accommodating changes after feasibility analysis.
- c. the system testing.
- d. the maintenance of system.

Answer: a

39. Which of the following is not included in waterfall model ?

- a. Requirement analysis
- b. Risk analysis
- c. Design
- d. Coding

Answer: b

40. Software Engineering is an engineering discipline that is concerned with:

- a. how computer systems work
- b. theories and methods that underlie computers and software systems
- c. all aspects of software production
- d. all aspects of computer-based systems development, including hardware, software and process engineering

Answer: c

41. Which process model is also called as classic life cycle model?

- a. Waterfall model
- b. RAD model
- c. Prototyping model
- d. Incremental model

Answer: a

42. Requirements prioritization and negotiation belongs to :

- a. Requirements validation
- b. Requirements elicitation
- c. Feasibility study
- d. Requirements reviews

Answer: b

43. Select the most appropriate statement about software engineering.

- A). Has been around as a discipline since the early 50's
- B). Is a set of rules about developing software products
- C). None of these choices
- D). Is an engineering discipline concerned with all the aspects of software production

Answer: D

44. Who proposed the spiral model?

- a) Barry Boehm
- b) Pressman
- c) Royce
- d) IBM

Answer: a

45. Which of the following document contains the user system requirements?

- a) SRD
- b) DDD
- c) SDD
- d) SRS

Answer: d

46. Software engineering is intended to support _____ software development.

- a) professional
- b) individual programming
- c) personal programming
- d) unstructured

Answer: a

47. Good software should deliver the required _____ and _____ to the user and should be maintainable, dependable, and usable.

- a) non-functionality and performance
- b) functionality and performance
- c) unreliable and underperformance
- d) functionality and non-availability

Answer: b

48. _____, where customers and engineers define the software that is to be produced and the constraints on its operation.

- a) Software development
- b) Software validation
- c) Software specification
- d) Software evolution

Answer: c

49. The waterfall model is an example of a _____ process—in principle, you must plan and schedule all of the process activities before starting work on them.

- a) incremental
- b) non plan-driven
- c) unstructured
- d) plan-driven

Answer: d

50. By developing the software _____, it is cheaper and easier to make changes in the software as it is being developed.

- a) incrementally
- b) non concurrently
- c) structured
- d) plan-driven

Answer: a

Unit II

1. Aggregation represents _____.

- a. is_a relationship
- b. part_of relationship
- c. composed_of relationship
- d. none of above

Answer: c

2. The feature of the object oriented paradigm which helps code reuse is _____.

- a. Object

- b. Class
- c. Inheritance
- d. Aggregation.

Answer: c

3. Which among these are the rules to be considered to form Class diagrams?

- a) Class symbols must have at least a name compartment
- b) Compartment can be in random order
- c) Attributes and operations can be listed at any suitable place
- d) Class symbols must not have at least a name compartment

Answer: a

4. Object diagram is used to show the design _____ view of a system.

- a. static
- b. dynamic
- c. logical
- d. process

Answer: a

5. Which of the following diagram is used to model the vocabulary of a system?

- a. Object Diagram
- b. Activity Diagram
- c. Class diagram
- d. Interaction Diagram

Answer: c

6. The object-oriented development life cycle is which of the following?

- a. Analysis, design, and implementation steps in the given order and using the steps no more than one time.
- b. Analysis, design, and implementation steps in any order and using the steps no more than one time.
- c. Analysis, design, and implementation steps in any order and using multiple iterations.
- d. Analysis, design, and implementation steps in the given order and using multiple iterations.

Answer: d

7. Composition is a stronger form of which of the following?

- a. Aggregation
- b. Inheritance
- c. Encapsulation
- d. Generalization

Answer: a

8. Sequence of messages is emphasized by _____ diagram.

- a. state chart
- b. sequence
- c. activity

d. collaboration

Answer: b

9. Which model in system modeling depicts the dynamic behavior of the system ?

- a. Context Model
- b. Behavioral Model
- c. Data Model
- d. Object Model

Answer: b

10. The UML supports event-based modeling using _____ diagrams.

- a. Deployment
- b. Collaboration
- c. State
- d. Class

Answer: c

11. _____ allows us to infer that different members of classes have some common characteristics.

- a. Realization
- b. Aggregation
- c. Generalization
- d. dependency

Answer: c

12. _____ & _____ diagrams of UML represent Interaction modeling.

- a. Use Case, Sequence
- b. Class, Object
- c. Activity, State Chart
- d. Class, State

Answer: a

13. Actual programming of software code is done during the _____ step in the SDLC.

- a. Maintenance and Evaluation
- b. Design
- c. Analysis
- d. Development

Answer: d

14. The context diagram is also known as _____

- a. Level-0 DFD
- b. Level-1 DFD
- c. Level-2 DFD
- d. Level-3 DFD

Answer: a

15. How is generalization implemented in Object Oriented programming languages?

- a. Inheritance
- b. Polymorphism
- c. Encapsulation
- d. Abstract Classes

Answer: a

16. In Design phase, which is the primary area of concern ?

- a) Architecture
- b) Data
- c) Interface
- d) All of the choices are correct

Answer: d

17. Which of the following describes "Is-a-Relationship" ?

- a) Aggregation
- b) Inheritance
- c) Dependency
- d) Composition

Answer: b

18. The state diagram

- a. depicts relationships between data objects
- b. depicts functions that transform the data flow
- c. indicates how data are transformed by the system
- d. indicates system reactions to external events

Answer: d

19. Grady Booch, James Rumbaugh, and Ivar Jacobson combined the best features of their individual object-oriented analysis into a new method for object oriented design known as _____?

- A. HTML
- B. XML
- C. UML
- D. SGML

Answer: C

20. What encapsulates both data and data manipulation functions ?

- A. Object
- B. Class
- C. Super Class
- D. Sub Class

Answer: A

21. Which of the following is a complementary approach to function-oriented approach ?

- A. Object oriented analysis
- B. Object oriented design
- C. Structured approach
- D. Both Object oriented analysis and design

Answer:D

22. Which of the following is delivered at the end of the Sprint?

- a) A document containing test cases for the current sprint
- b) An architectural design of the solution
- c) An increment of done software
- d) Wireframes designs for User Interface

Answer:c

23. Select the option that suits the Manifesto for Agile Software Development

- a) Individuals and interactions
- b) Working software
- c) Customer collaboration
- d) All of the mentioned choices are correct

Answer:d

24. Agile Software Development is based on

- a) Incremental Development
- b) Iterative Development
- c) Linear Development
- d) Both Incremental and Iterative Development

Answer:d

25. How many phases are there in Scrum ?

- a) Two
- b) Three
- c) Four
- d) Scrum is an agile method which means it does not have phases

Answer: b

26. According to Agile manifesto what carries more value?

- A. Individuals and interactions over processes and tools.
- B. Individuals and interactions over people and technique.
- C. Individuals and interactions over projects and tools.
- D. Individuals and interactions over products and tools.

Answer: A

27. An Agile customer ...

- A. Must have a good understanding of Agile techniques
- B. Will always receive worse systems than their non-Agile counterparts
- C. Will typically get business value delivered early
- D. Will need to work out at the gym regularly

Answer: C

28. The Agile approach to documentation is:

- A. Do no documentation – it is a waste of time
- B. Do plenty of documentation, to prove you have done a good job
- C. Do the necessary documentation to support the use of the product
- D. Do more documentation than usual, because Agile is risky

Answer: C

29. The working culture of an Agile team is ...

- A. Collective
- B. Collaborative
- C. Connective
- D. Contemplative

Answer: B

30. An Agile approach values ...

- A. Process over people
- B. People over process
- C. Plans over process
- D. Plans over people

Answer: B

31. Who is responsible for sprint meeting?

- a) Product owner

- b) Scrum team
- c) Scrum master
- d) Architect

Answer: c

32. Scrum is iterative. The iteration is called

- A) sprite
- B) IT1
- C) sprint
- D) sprinte

Answer: C

33. Select one of the following options that explains the meaning of daily stand-up in agile?

- A) It is the meeting to ask future work schedule
- B) A meeting where the team is asked the things done on the previous day and the plan for the current day along with the problems they faced during these tasks
- C) A stand-up lecture
- D) no such meetings

Answer: B

34. User requirements are expressed as _____ in Extreme Programming.

- a) implementation tasks
- b) functionalities
- c) scenarios
- d) algorithms

Answer: c

35. XP follows Test _____ Development practice.

- a) first
- b) last
- c) after
- d) none of these choices

Answer: a

36. _____ is a XP Practice?

- a) Pair Programming
- b) Individual Coding
- c) Plan intensive
- d) none of these choices

Answer: a

37. Agile model advocates which one of the following approaches?

- a. In each increment, get something “quick and dirty” delivered and thereby save time
- b. In each increment, get something simple released as quickly as possible

- c. In each increment, get something of business-value delivered as quickly as possible, consistent with the right level of quality
- d. In each increment, get something delivered once it has been fully documented and the documentation has been signed off as complete

Answer: c

38. Pair programming intends to take which one of the following best practices to the extreme?

- a) Design
- b) Incremental development
- c) Code review
- d) Simplicity

Answer: c

39. A user story is

- a. A series of interactions by the user (Actor) with the system and the response of the system.
- b. A narrative text(s) that describe an interaction of the user with the value of from the system.
- c. A narrative text(s) that describe an interaction of the user and the system, focusing on the value a user gains from the system.
- d. Non descriptive

Answer: c

40) System Testing is a ____.

- a. Black box testing
- b. White box testing
- c. Grey box testing
- d. Both a and b

Answer: a

41) Verifying that whether individual software components are functioning correctly and identifying the defects in them is objective of which level of testing?

- a. Integration Testing
- b. Acceptance Testing
- c. Unit Testing
- d. System Testing

Answer: c

42. Which of the following is non-functional testing?

- a) Black box testing
- b) Performance testing
- c) Unit testing
- d) None of the mentioned

Answer: b

43. Which traditional order in Software Testing is organized?

- a. Integration Testing

- b. System Testing
- c. Unit Testing
- d. Acceptance Testing

- a) a, d, c, b
- b) b, d, a, c
- c) c, a, b, d
- d) d, b, c, a

Answer: c

44. Verification and Validation is focused on _____ respectively.

- a. Product
- b. Process and Product
- c. Both
- d. None

Answer: b

45. Equivalence Partitioning comes under which type of Testing?

- a. White Box Testing
- b. Green Box Testing
- c. Grey Box Testing
- d. black box testing

Answer: d

46. _____ and _____ processes are concerned with checking that software being developed meets its specification and delivers the functionality expected by the people paying for the software.

- a) Verification and validation
- b) Analysis and Design
- c) code and design
- d) analysis and code

Answer: a

47. In _____ testing, where a separate testing team tests a complete version of the system before it is released to users.

- a) User
- b) Release
- c) Development
- d) Unit

Answer: b

48. The _____ should show that, when used as expected, the component that you are testing does what it is supposed to do.

- a) software
- b) hardware
- c) Development
- d) test cases

Answer: d

49. _____ tests have to be designed to ensure that the system can process its intended load.

- a) Performance
- b) hardware
- c) system
- d) user

Answer: a

50. In _____ testing, where customers test a system to decide whether or not it is ready to be accepted from the system developers and deployed in the customer environment.

- a) unit
- b) component
- c) system
- d) Acceptance

Answer: d